

Spotify Song Features Case Study Hook

A Case Study by Stephanie Diau

Topic:

Time Series Analysis of Song Features from Spotify Data

Context:

There is a common belief that certain songs are more popular at different times of the year, for example, you may associate slower, “sadder” songs with the winter months while faster, upbeat songs are associated with the summer months. This isn’t difficult to believe, terms like “seasonal depression” suggest that we might be more drawn to specific types of songs given their mood and time of year.

This idea frequently occurs in pop culture as well. You might see up-and-coming artists attempt to advertise their music as the next “song of the summer” or hear people say “She’s defrosting..” in reference to how Mariah Carey’s hit single “All I Want For Christmas Is You” reappears on the charts every time the weather gets colder. From a marketing standpoint, music labels and artists may consider the time of year as well when releasing new music. Seasonality is a well-established factor in consumer behavior, and the music industry is certainly no exception.

But are these assumptions supported by data analysis, or is it just a cultural perception?

Motivation:

You are a data scientist for a music streaming platform like Spotify. You and your team are interested in exploring how modern songs evolve over time, specifically finding if seasonal patterns exist in song features based on their month of release. Your role is to use time series analysis methods to determine if there is significant evidence of seasonality from songs released between the years of 2000 and 2020.

Identifying the existence of (or lack thereof) patterns can assist in song marketing strategies, improve music recommendation algorithms, and enhance platform-made playlists. The key song features of interest include: Acousticness, Speechiness, Valence, Tempo, and Duration.

Deliverable:

From the provided GitHub repository, you will have access to cleaned, filtered data, analysis scripts, and a rubric for this case study. Your final deliverable will include a written report based on your experience and findings, as well as your own repository containing the materials that you used. You will submit only the link to the repository as the PDF of your report will be included within.